



Subject: Programming with Java

ASSIGNMENT No : 9

TITLE : Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Outcome

Exercise

1. Create a **Bank Account** program where account number is final and cannot be changed.

```
AS9EX1.java 1 | ASS5EX2.java 3 | AS6EX1.java 2 | ShapeDer | AS7EX2.java
J AS9EX1.java > BankAccount
1 class BankAccount {
2
3     final int accountNumber = 12345;
4     String accountHolder = "Rahul";
5     double balance = 5000.0;
6
7     void display() {
8         System.out.println("Account Number: " + accountNumber);
9         System.out.println("Account Holder: " + accountHolder);
10        System.out.println("Balance: " + balance);
11    }
12
13    Run main | Debug main
14    public static void main(String[] args) {
15        BankAccount b = new BankAccount();
16        b.display();
17    }
```

OUTPUT-

```
PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Account Number: 12345
Account Holder: Rahul
Balance: 5000.0
```

2. Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
X1.java 1 J ASS5EX2.java 3 J AS6EX1.java 2 J ShapeDer :: || ↺ ⬇ ⬆ ↻
J AS9EX1.java > LibraryBook > display()
1 class LibraryBook {
2
3     final String ISBN = "978-81-12345-67-8";
4     String title = "Java Programming";
5     String author = "James Gosling";
6     double price = 450.0;
7
8     void display() {
9         System.out.println("ISBN: " + ISBN);
10        System.out.println("Title: " + title);
11        System.out.println("Author: " + author);
12        System.out.println("Price: Rs." + price);
13    }
14
15    Run main | Debug main
16    public static void main(String[] args) {
17        LibraryBook book = new LibraryBook();
18        book.display();
19    }
```

OUTCOME-

```
PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS
ISBN: 978-81-12345-67-8
Title: Java Programming
Author: James Gosling
Price: Rs.450.0
```

Conclusion

The program was successfully implemented to demonstrate the use of the **final** keyword with variables, methods, and classes in Java. It showed how the **final** keyword prevents

reassignment of variables, overriding of methods, and inheritance of classes, thereby improving program security and reliability. This practical enhanced the understanding of data protection and code stability using the **final** keyword in object-oriented programming.